

The Pathfinder engine

github.com/vd1/pathfinder

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Abstract

Pathfinder is an “AI for science” protocol aiming at creating new research ideas at scale by searching the scientific literature. Its engine scans and scores pairs of papers from selected corpora for potential connexions. Top scoring pairs are then given to AI agents to produce a research note, which is then submitted to an agent verifier who decides whether to elaborate the note into a publication output for external assessment. No human is required during the process once the initial corpora is defined.¹

The purpose of this note is to describe the Pathfinder engine as a state transition system at a level of details suitable for human and AI readers, so they can jointly and easily re-implement it, should they wish to, and adjust it to their specific needs.

1 The Pathfinder program

Pathfinder takes two corpora of scientific papers, concretely arXiv deposits, Q and P , and, using AI agents (currently using OpenAI and Anthropic models), looks for pairs $(q, p) \in Q \times P$ with a scientific connexion which seems interesting enough to try to elaborate it as a publishable note. Ideally, the final output satisfies the usual constraints of correctness, novelty, relevance, readability, and exposition of related work.

Note that Pathfinder is primarily concerned with finding and exploring connexions. In the course of exploring a connexion agents may state and prove propositions, but one is not given in advance a conjecture to prove/dis-prove. The idea of generating new ideas by exploring connexions is not new. Swanson’s *undiscovered public knowledge* is a direct precedent for this motivation: findings may already be published in

¹ Eventually, in a world where agents dominate the scientific throughput, a publication-looking output may not be the better format; but that seems appropriate at the moment for a hybrid AI/human led research process.

separate literatures, yet what they imply together remains unnoticed until someone brings them into contact [1]. The missing step is not necessarily a new observation, but recognising and investigating a connexion between existing findings. Pathfinder seeks to make that search systematic.

Gentner’s account of analogy describes a more specific way in which a connexion can work [2]. Two settings may involve very different objects but share a pattern of relationships. Recognising that pattern can suggest transferring an argument or mechanism from one setting to the other; similar terminology alone is not enough. Analogy is therefore a possible route to a Pathfinder seed, not a description of every seed. A paper may instead provide a direct application of another’s theorem, expose a failure of its assumptions, or suggest how to repair its formulation. Swanson motivates the broader search for knowledge between literatures; Gentner helps explain a particular form of reasoning within it. In each case, the research phase must establish what the proposed connexion actually supports.

Closer computational precedents include Krenn and Zeilinger’s *SemNet* [3], which suggests new links between scientific concepts in a network extracted from the literature, and Gu and Krenn’s *SciMuse* [4], which combines a knowledge graph with a language model to propose research ideas, evaluated by experts for interest. Pathfinder starts instead from pairs of papers and asks what might follow from bringing their specific claims or methods together. Its aim is also to investigate the proposed connexion, not only suggest it: the research phase may establish a conditional result, expose an obstruction, or conclude that further evidence is needed. The distinction is therefore in the starting unit and the subsequent investigation, not in the general idea of using AI to find scientific connexions.

What is new is that AI agents are now available that can plausibly stand in for human experts and search this space at scale. Whether the systematic search for connexions leads to useful scientific knowledge can now be tested. Pathfinder is a first step in this direction.

The goal of this note is to specify the underlying protocol at a level sufficient for an agent or a human programmer to reimplement it, without prescribing every detail of its present implementation.

How do human experts get new ideas? It happens sometimes through seminars. The seed of an idea is provided by someone in the audience, and a connexion made, which, in favourable circumstances, can lead to a publication. By using modern agents (which are competent experts on every corpus, and always “willing” to contribute), one can do it faster, cheaper, and, therefore, at a larger scale. We have run Pathfinder on sets of papers of size about a hundred each for a reasonable token budget (for budget-related considerations, see Appendix A).

Another comparison is with word embeddings and the idea that what a word means can be seen from the company it keeps. A scientific paper can likewise be understood through other papers to which it can be related: work that may use its methods, extend its results, or challenge its assumptions. Citations record the past connexions. Pathfinder looks at possible ones. Its scan proposes links that the citation graph does not show; its research phase tries to establish what those links mean and whether they yield anything useful.

2 The 10-by-10 experiment

To test the engine on a small scale, we picked the ten most recent papers (at the time of query, Sep 15, 2026) returned by arxiv for the queries “mechanism design” (for Q) and “agentic cooperation” (for P). These papers are labelled Q1, ..., Q10 and P1, ..., P10. The Q-collection includes economic mechanisms, learning-augmented algorithms, robotic hardware, and astronomical instrumentation. The P one includes multi-agent learning, perception, distributed optimisation, and energy systems.²

The scanner agent (Sonnet 5) scored each of the 100 pairs from 0 to 100 for feasibility and scientific-gain, and proposed a seed connexion. It did so being only given titles and abstracts (thereby keeping contexts small). The ranking score is the product. We picked a threshold of 1500 eliciting 13 pairs for the research phase. The shortlist also retained the existing Q1P2 investigation, whose score was 600, giving fourteen research threads. (Because it looked promising to the human in charge.)

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
Q1	1925	600*	300	150	75	1500*	0	75	1050	1400
Q2	0	0	0	0	0	0	0	0	300	0
Q3	200	0	1925	0	75	1575	0	0	3300	3000
Q4	300	0	2100	0	25	3150*	0	25	1800	3575
Q5	75	0	0	9	0	0	0	0	0	0
Q6	25	0	0	0	100	50	0	0	600	240
Q7	2200*	0	0	120	75	150	1650*	0	150	25
Q8	0	25	0	0	0	200	0	1650	0	50
Q9	0	0	0	0	0	0	0	0	0	0
Q10	0	0	0	0	0	0	0	0	0	0

0 scan score: feasibility $\times$ gain 3575

Figure 1 100 pairs scanned, 14 retained for research (black outline), 5 accepted as papers (starred). Numbers are the combined feasibility and gain score.

The campaign used several models (Table 1). Totals include retries and failed calls, not just successful transitions. Recorded dollar costs are indicative calculated accounting values. Appendix A discusses per-stage costs and elapsed times, their relation to token usage, and an economic comparison with human costs.

2.1 Outcomes

Table 2 gives the outcomes.

The highest-scoring pair, Q4P10, did not reach DRAFT: a generic incentive inequality was not enough to establish the proposed connexion to the energy simulation.

² The question of how to choose the corpora is not discussed in this note.

Model recorded in receipts	Calls	Recorded cost (USD)
claude-sonnet-5	106	3.34
claude-opus-5	30	77.22
gpt-5.6-sol	194	106.02
Total	330	186.58

Table 1 Root-campaign model usage.

Campaign stage or outcome	Number
Candidate pairs	100
Investigated pairs	14
Research outcomes: PAUSE	8
Research outcomes: PAUSE-ON-ITERATE	1
Research outcomes: DRAFT	5
Accepted manuscripts	5

Table 2 The pilot’s selection and output counts. Rows describe successive stages or different output branches.

Conversely, the retained low-scoring Q1P2 did produce an accepted manuscript (a potential false negative for the scanning phase). The scan ranking was not a reliable ordering of the eventual outcomes in this small sample. Other paused threads lacked a controlled experiment, an executable interface, or evidence that the proposed transfer added something specific to the pair. For example, Q3P10 required unavailable market experiments. For Q3P3, the researchers identified a missing comparison in P3: does adversarial review outperform repeated review by a single agent given the same computational budget? They proposed this control, but lacked the task instances and execution setup needed to reproduce and extend P3’s experiments. A pause therefore need not mean that the underlying scientific idea is false.

The accepted outputs are summarised in Table 3. Their strongest contributions are narrower than their initial ambitions: an interface-specific guarantee, an accounting repair, an identification construction, or a conditional perturbation bound. The following excerpts present the mathematical content in condensed notation rather than reproduce the manuscripts. Their stated assumptions are part of the results.

2.2 Example 1: Q4P6

Consider the pair:

q [arXiv:2609.01595](https://arxiv.org/abs/2609.01595), *Mechanism Design for Alignment and Control*, studies how to reward agents so that they have an interest in reporting what they know truthfully.

p [arXiv:2607.27967](#), MARS-RA: Rank Aggregation for Credit Assignment via Multimodal Comparisons in Embodied Multi-Agent Cooperation, uses an external AI model to look at images from cooperating agents and compare who contributed more. These comparisons are combined into scores used to guide the agents' training.

The seed asks: in *p* instead of asking an external model who contributed more, could we ask the cooperating agents themselves, and use *q*'s mechanism to make it worthwhile for them to tell the truth?

This suggests a change to *p*: instead of relying on an external judge, whose comparisons may be noisy, rely on the participants themselves. Clearly, asking the participants introduces a reason to misreport if they benefit from receiving more credit. Under its assumptions, *q* offers a way to make truthful reporting worthwhile. However, a second difficulty remains. An agent can tell the truth about what it saw and still have an incomplete or misleading view of who helped the team. Will the agents tell us what they know, and is what they know enough to rank contributions? These are separate questions.

The research phase reached a positive, conditional conclusion: *an external judge is unnecessary if the peers' observations satisfy explicit conditions. Their reports alone then determine the ranking specified by the comparison model, without anyone supplying the correct answers to individual comparisons.*

The idea is to use agreement between observers to work out how reliable each is. Suppose truthful observers are better than chance, have stable error rates that are the same whichever agent really did better, and make independent errors once the underlying comparison is fixed. Their observations must overlap: for example, a triangle of observers repeatedly judging shared episodes provides a starting point for

Pair	Source connexion	Accepted manuscript's main result
Q1P2	Learning-augmented algorithms and concept-guided exploration	Room-level calibration certifies the non-abstained decisions of a first wall-distance gate.
Q1P6	Learning-augmented algorithms and MARS-RA credit assignment	A fixed-population reward ledger repairs potential shaping under changing coalitions.
Q4P6	Mechanism design for alignment and MARS-RA credit assignment	A conditional interface recovers contribution rankings from noisy peer signals.
Q7P1	Stochastic-stability mechanism design and distributed CVaR optimisation	A uniform surrogate-error bound controls static welfare loss and conditional participation loss.
Q7P7	Stochastic-stability mechanism design and carbon-response forecasting	A voluntary workload-migration formulation and a directional forecast-error certificate.

Table 3 Scientific content of the internally accepted root-campaign papers. Acceptance records a workflow judgement, not established novelty or validity.

calibration. The other observers can be calibrated through further overlaps. We can then correct for their errors and recover the underlying comparison probabilities. Under the assumed statistical ranking model, these give a common ranking, provided the comparisons connect all the agents being ranked. Thus truthful but imperfect reports can collectively suffice, even when no observer knows all the answers.

The argument uses an established calibration method; its contribution here is to identify a setting in which p 's external judge could be replaced. The result assumes exact agreement frequencies, so it does not yet say how much data would be needed in practice. For strategic reporters, q 's incentives also require information about what agents expect their peers to report, which the calibration does not automatically supply. Finally, the recovered ranking measures actual contribution only if the comparison model captures it. The conclusion is therefore a judge-free ranking result under stated information conditions, not yet a complete mechanism for making those conditions hold.

2.3 Example 2: Q7P1

Consider another example:

q [arXiv:2608.29130](#), *A Systematic Approach to Mechanism Design with Stochastic Dynamic Stability*, proposes payments and a learning procedure intended to make self-interested agents share limited resources efficiently.

p [arXiv:2609.04460](#), *Distributed risk-averse optimization via CVaR*, uses noisy loss observations to optimise conditional value-at-risk (CVaR): the average loss in a chosen worst fraction of cases, rather than over all cases.

The seed asks: could we make q 's allocation scheme risk-averse, using p 's way of learning from losses, while keeping its guarantees?

The researcher team first found that the algorithms do not combine directly: q requires solving a local optimisation problem, whereas p only estimates a direction of improvement. The first investigation therefore asked what could be guaranteed about p 's approximation to risk, independently of the learning procedure.

Its paper establishes that *minimising the approximate risk gives near-optimal true risk over the same feasible choices*, with an explicit error bound decreasing with larger sample batches and less smoothing. This concerns the approximation averaged over sampling, not each realised batch. The paper also bounds possible losses from participation, *conditional on a suitable mechanism existing*; it does not prove that premise.

A separate rerun exposed problems in q , not p , before introducing risk aversion. Its stability and implementation conditions are incompatible, and its printed payments can leave a participant receiving less than what it pays out, even at equilibrium. The first paper's approximation bound survives because it does not depend on those payments or conditions. Its mechanism application, however, remains unestablished: q as printed does not justify the guarantees that application needs.

The rerun’s subsequent paper follow-up repairs the payments by adding a term depending only on other agents’ messages, leaving each agent’s best choices unchanged. Under stronger curvature and feasibility assumptions, *the corrected mechanism implements the unique optimum of a fixed risk approximation, with balanced payments and no participation loss at equilibrium under that approximation*. Error bounds control the shortfalls when assessed using true risk. The objective sums individual risks, rather than measuring the risk of combined losses. A changed learning procedure also has a convergence proof under further assumptions; efficient learning remains open.

These are not merely different, equally satisfactory answers to the seed. The first run supplies useful error control, but leaves a crucial mechanism premise unresolved. The later investigation exposes that weakness and develops a repair using the same approximation approach. The progression illustrates both the value of revisiting a connexion and a limit of internal acceptance: a useful result may survive while its intended application still needs substantial work.

2.4 A selective perception guarantee: Q1P2

q [arXiv:2609.04787](#), *Learning-Augmented Algorithms: Guarantees, Construction Mechanisms, and System-Level Implications*, surveys algorithms that use predictions while retaining guarantees when those predictions are wrong.

p [arXiv:2608.23650](#), *Concept-Guided Exploration: Building Persistent, Actionable Scene Graphs*, lets a robot build a map of rooms and doors. Its estimated room walls guide where it looks for doors.

The seed asks: could *q*’s approach to fallible predictions give *p*’s door detector a guarantee when the estimated walls are wrong?

This pair also illustrates how the scanner can overlook a promising connexion. Sonnet 5 assigned feasibility 30 and gain 20, giving 600, below the shortlist threshold of 1500. On 21 September, a fresh Astra instance (gpt-6-astra) received only the same titles and abstracts and the current scanning rubric, without the earlier score, seed or research outcome. It assigned feasibility 64 and gain 58, giving 3712: the pair would have passed the threshold. Astra proposed combining the robot’s room-based predictions with a fallback exploration strategy, then asking whether correct predictions save work while incorrect ones cause only a bounded extra cost. It also identified a difficulty specific to this setting: exploration changes the information on which subsequent predictions depend.

In this instance, Astra recognised substantially more promise in the connexion. Its proposal was broader than the selective perception guarantee developed below; it did not independently recover or validate that result. The rescore is an encouraging

example of recovering a candidate missed by the original threshold, rather than evidence of generally superior detection from a single comparison.³

Concept-guided exploration uses estimated walls to decide which scan points to retain for subsequent door detection. Its first gate accepts a point z when its distance from the wall set is at most $\delta = 0.15$ metres. The problem is discontinuity: a small wall error can change a binary decision. The paired reading replaces an unrestricted robustness claim with a selective guarantee for this particular interface.

Let $\widehat{W}$ and W^* be the estimated and reference wall sets, assumed nonempty and compact. Their Hausdorff distance controls every point's distance error:

$$d_H(\widehat{W}, W^*) \leq \varepsilon \implies |\text{dist}(z, \widehat{W}) - \text{dist}(z, W^*)| \leq \varepsilon \quad \text{for every } z. \quad (1)$$

Freeze the room-fitting pipeline independently of calibration. From n exchangeable calibration episodes, form Hausdorff errors $R_1, \dots, R_n$ and take ε_α to be their $\lceil (n+1)(1-\alpha) \rceil$ th order statistic, or $+\infty$ when that index exceeds n . In an exchangeable new episode, accept a point only if $\text{dist}(z, \widehat{W}) + \varepsilon_\alpha \leq \delta$, reject it only if $\text{dist}(z, \widehat{W}) - \varepsilon_\alpha > \delta$, and otherwise abstain. The resulting statement is

$$\Pr\{\text{all non-abstained first-gate decisions match the reference gate}\} \geq 1 - \alpha. \quad (2)$$

The useful step is that a single room-level coverage event certifies all these decisions, without a union bound over scan points. Conformal calibration and the distance inequality are established tools; the proposed contribution is their transfer to this detector interface. The guarantee is marginal across rooms, permits arbitrarily frequent abstention, and says nothing by itself about final doors, map topology, or robot motion. No episode-level calibration data were available, so practical usefulness remains unmeasured. The output is a theorem and a validation protocol, not a demonstrated safer robot.

2.5 An accounting repair for changing coalitions: Q1P6

q [arXiv:2609.04787](#), *Learning-Augmented Algorithms: Guarantees, Construction Mechanisms, and System-Level Implications*, surveys how algorithms can benefit from predictions without losing their guarantees when the predictions are inaccurate.

p [arXiv:2607.27967](#), *MARS-RA: Rank Aggregation for Credit Assignment via Multimodal Comparisons in Embodied Multi-Agent Cooperation*, uses an outside AI judge to compare cooperating agents' contributions. The resulting scores adjust their training rewards, including when agents join or leave the active team.

The seed asks: could *q*'s treatment of prediction errors explain how mistakes in *p*'s comparisons affect its reward guarantees?

MARS-RA turns pairwise comparisons into contribution scores and then into potential-based reward shaping. Its active coalition can change, so consecutive

³ The rescore used a fresh agent instructed to use no tools, not the original scanner transport. The current prompt was not checked against a historical snapshot. Inputs, response and provenance are saved in `experiments/2026-09-21-174126-astra-blind-Q1P2/`.

potential vectors need not have the same index set. The investigation found that specifying prediction accuracy was premature: the reward accounting first needed a coherent definition.

The repair embeds active agents' softmax credits in a fixed population, assigns zero potential to inactive coordinates, and maintains a reward ledger for every population member on every transition. For agent i , use the shaping payment $F_t^i = \gamma \Phi_{t+1}^i - \Phi_t^i$, with terminal potential $\Phi_T^i = 0$. For finite episodes and a fixed initial augmented state,

$$\sum_{t=0}^{T-1} \gamma^t F_t^i = -\Phi_0^i + \gamma^T \Phi_T^i = -\Phi_0^i. \quad (3)$$

The payoff shift is policy-independent, preserving unilateral payoff differences and best responses, regardless of the comparison outputs. This telescoping principle is prior work; the contribution is the explicit population and transition accounting needed to apply it here.

In particular, paying an agent only on transitions whose starting state has that agent active drops its entry payment. For a participation interval $a, \dots, b-1$, with zero potential before entry and after exit, the recorded sum becomes

$$\sum_{t=a}^{b-1} \gamma^t F_t^i = -\gamma^a \Phi_a^i, \quad (4)$$

rather than the zero sum obtained when both boundary payments are included. That term can depend on policies through entry timing and state; on re-entry it can also depend on the returning agent's earlier actions. The paper distinguishes this from conventions that omit the exit payment as well. It does not claim that MARS-RA's implementation actually uses either mask, or that the repair improves finite-training performance. The scientific gain is a diagnosed ambiguity and a sufficient repair, not a new general reward-shaping theorem.

2.6 From peer signals to rankings: Q4P6

q [arXiv:2609.01595](#), *Mechanism Design for Alignment and Control*, studies rewards that give agents an interest in reporting what they know truthfully.

p [arXiv:2607.27967](#), *MARS-RA: Rank Aggregation for Credit Assignment via Multimodal Comparisons in Embodied Multi-Agent Cooperation*, uses an outside AI judge to compare agents' contributions from images, then combines those comparisons into scores for training.

The seed asks: could the participants replace the outside judge, with *q*'s incentives making their reports truthful and their reports supplying *p*'s contribution ranking?

This pair asks whether truthful peer information can feed a contribution ranking, rather than merely resemble a pairwise preference model. The accepted result supplies a conditional statistical interface. On comparison edge $e = (i, j)$, let the latent sign $X_e \in \{-1, 1\}$ satisfy $\Pr(X_e = 1) = 1/(1 + \exp[-(c_i - c_j)])$. Observer r reports

$Y_{er} \in \{-1, 1\}$, with $\mathbb{E}[Y_{er} \mid X_e] = a_r X_e$ and $0 < a_r \leq 1$. Reports are conditionally independent given the same latent draw, and reliabilities are stable across the edges being calibrated.

Population agreement moments then satisfy $q_{rs} = \mathbb{E}[Y_{er} Y_{es}] = a_r a_s$. A triangle of overlapping observers identifies its positive reliabilities, for example

$$a_r = \sqrt{\frac{q_{rs} q_{ru}}{q_{su}}}, \quad p_e = \frac{1}{2} \left(1 + \frac{\mathbb{E}[Y_{er}]}{a_r} \right), \quad c_i - c_j = \log \frac{p_e}{1 - p_e}. \quad (5)$$

Calibration can propagate through observer overlaps; a connected comparison graph then identifies the scores up to a common translation. The triangle is a sufficient calibration pattern, not a necessary one.

The calibration algebra belongs to established noisy-label models. What the investigation contributes is an explicit interface between peer reports and the ranking stage, together with its limitations. Incentives do not follow from identifiability: the mechanism-design source requires additional information about conditional peer beliefs to make truthful reporting an equilibrium. Nor do correlations from independently redrawn latent comparisons instantiate the shared-draw moment equation. Finite-sample guarantees, strategic collusion, and changing observer populations remain outside the result. Recovering latent comparison scores also does not by itself identify a semantic contribution measure such as Shapley value.

The connexion can be credited with posing the interface question and exposing these conditions, rather than with inventing the calibration identity. Making truthfulness and calibration jointly achievable remains a substantive open step. A later targeted Astra reassessment retained the central DRAFT claim while requiring local corrections; it did not establish publication readiness.

2.7 A static risk bound, and what subsequent scrutiny changed

q [arXiv:2608.29130](#), *A Systematic Approach to Mechanism Design with Stochastic Dynamic Stability*, proposes messages, payments and a learning procedure intended to make self-interested participants share limited resources efficiently.

p [arXiv:2609.04460](#), *Distributed risk-averse optimization via CVaR*, learns from noisy observations while paying particular attention to losses in the worst fraction of cases.

The seed asks: could *p*'s way of learning about risk make *q*'s resource-sharing scheme risk-averse while preserving its guarantees?

The original Q7P1 proposal sought to combine distributed risk-averse optimisation with a mechanism possessing strategic and dynamic guarantees. The accepted root-campaign paper established a more limited statement. Let C_n be an agent's population CVaR cost at tail level α_n and $\tilde{C}_n$ its expected empirical, spatially smoothed surrogate. Under the paper's bounded-loss and samplewise Lipschitz assumptions, on a common admissible domain,

$$\sup_x |\tilde{C}_n(x) - C_n(x)| \leq b_n, \quad b_n = L_n \delta_n + \frac{U_n}{\alpha_n} \sqrt{\frac{2\pi}{s_n}}, \quad (6)$$

where L_n bounds sensitivity, δ_n is the smoothing radius, U_n bounds absolute loss, and s_n is the sample size. If x^* and $\tilde{x}$ minimise the true and surrogate aggregate costs over the same feasible set, respectively, this yields

$$\sum_n C_n(\tilde{x}) - \sum_n C_n(x^*) \leq 2 \sum_n b_n. \quad (7)$$

The result controls static surrogate error. It does not prove that the source mechanism's matrix certificate is feasible, that the two papers' learning dynamics can be interchanged, or that restricting the feasible domain for smoothing is harmless.

Subsequent Astra investigations, stored separately from the root campaign, show why those distinctions matter. The Q7P1 follow-up identified an obstruction to the mechanism source's strict matrix certificate and a counterexample to its explicit payment's participation guarantee. A direct source audit on 14 September confirmed these defects and a separate printed sign inconsistency in the learning formulation. The participation example refutes the explicit formula's claimed property, not a conditional theorem whose premises that formula fails. A proposed opponent-only transfer correction preserves best responses while restoring participation and equilibrium budget balance under explicit assumptions; a direct equilibrium argument recovers welfare-maximising allocations without demanding unique prices. An efficient stochastic implementation of the replacement learner and external assessment of the repair remain open. These are later outputs, not additional successes in the original fourteen-thread denominator.

2.8 A directional emissions guarantee: Q7P7

q [arXiv:2608.29130](#), *A Systematic Approach to Mechanism Design with Stochastic Dynamic Stability*, proposes payments and a learning procedure intended to coordinate participants who pursue their own interests.

p [arXiv:2607.26560](#), *Emission-Forecasting-Based Spatial-Temporal Carbon Response: A Multi-Agent Attention-Enhanced Deep Learning Framework*, forecasts the carbon intensity of electricity and moves flexible demand across locations and times to reduce emissions.

The seed asks: could *q*'s incentives preserve *p*'s emissions reductions when the owners of flexible demand act strategically rather than simply following a central schedule?

The accepted paper similarly contains a useful separation. For a fixed baseline and proposed migration schedule, write Δg for the change in nodal electrical withdrawals, $\hat{e}$ for forecast carbon intensities, and e for realised intensities. With the sign of Δg chosen so that $\hat{M} = \hat{e}^\top \Delta g$ is the forecast emissions reduction, its directional certificate is

$$M \geq \hat{M} - \|e - \hat{e}\|_* \|\Delta g\|. \quad (8)$$

A positive right-hand side certifies a reduction under the exogenous-intensity model. This application of the dual-norm inequality does not depend on the mechanism's stability certificate. By contrast, the paper's proposed implementation of voluntary

migration explicitly assumes that certificate. The later Q7P1 obstruction therefore raises a dependency concern for that conditional implementation claim too; this is an implication of the follow-up, not a completed reassessment of Q7P7.

2.9 What the pilot evaluates

The clearest positive observation is that some investigations moved beyond an analogy to a checkable mathematical statement, a source-level objection, or a repair with explicit assumptions. The negative and narrowed outcomes are also informative: attractive connexions often stopped at unavailable experiments, incompatible interfaces, or unverified premises. The saved ledgers and readable notes expose those stopping points to an expert reader.

Five accepted manuscripts out of fourteen investigations is a workflow yield, not a measured rate of new scientific discoveries. The study has no unassisted or single-agent control, no blinded external evaluation, no common frozen model configuration, and no systematic test of practical effect size. The corpus is small and query-dependent, the shortlist is selected, and some threads received additional human-directed work. Prior work searches support bounded novelty claims, not proof of originality. The later correction of a source dependency further shows that internal acceptance is revisable.

Accordingly, the pilot supports a limited conclusion: this protocol can produce substantive candidates for expert assessment, with evidence and limitations attached. Establishing how often those candidates are correct, new, and scientifically useful requires external assessment and a more controlled campaign. The specification that follows explains the process that generated these records; it should not be read as a proof that following the process guarantees a discovery.

Record provenance. The root-campaign counts come from `Q.jsonl`, `P.jsonl`, `scan.jsonl`, `shortlist.json`, and `receipts.jsonl`, together with each thread's saved status, manuscript-review record, and readable-note build record. The mathematical excerpts are drawn from the accepted manuscripts under `threads/`, not inferred from scan scores. The later Astra reassessment, paper follow-ups, and source audit are separate records under `experiments/`; neither their calls nor their revised manuscripts are included in the root-campaign totals above.

3 Labelled transition system

The diagrams below describe the control positions of a labelled transition system. A complete state also carries the working record and the resource counters. We write a transition as

$$(\sigma, M, b) \xrightarrow{\ell} (\sigma', M', b'),$$

where σ is the position, M the persistent record, b the counters and allowances, and ℓ the event or decision enabling the move. For example, a verifier's `ITERATE` label

leads back to research only when another research round is allowed. The diagrams suppress M and b ; the paragraphs give the guards and effects of each transition.

The agents propose scientific judgements. The runner validates their replies, records the results, and applies the transition rules. Its choice of the next position is mechanical, although the model response enabling that choice is not deterministic. The intended properties are bounded automatic work, decisions traceable to their evidence, and recovery from a saved position rather than restarting an investigation from nothing. Limits of these properties are made explicit below.

We use this notation throughout the specification. For the complete thread, the control position is a product

$$\sigma = (\rho, \pi, \eta; \omega),$$

where ρ is the research position, π the paper position, η the readable-account position, and $\omega = (\omega_R, \omega_P, \omega_A)$ records a `RUNNABLE`, `STOPPED`, or `BLOCKED` operational mode for each branch. The output positions initially equal `IDLE`. A notation such as $\sigma[\rho \leftarrow \text{VERIFY}]$ changes that component only. Thus the output branches can progress independently after research ends; `BLOCKED` is not an additional scientific research verdict.

Write $M \oplus z$ for the named record update z , including any required history entry. It may replace a current artifact as well as append to a history; it does not imply that every old artifact version is retained. The loop counters in b are r (research round), u (note repairs used), w (paper round), and e (account attempt), with caps $\bar{r}, \bar{u}, \bar{w}, \bar{e}$. Other components track call, time, and campaign allowances. For a completed model call, M_z, b_z denote its saved output and receipt and the corresponding accounting update; displayed counter assignments are additional to that update. Unmentioned components are unchanged. Guards refer to recorded quantities, not to unobserved model work. Each rule is a logical transition of the specification, not a claim that the reference implementation writes its files atomically.

4 The research phase

The research phase takes the incoming message (q, p, c) and ends in one of four terminal states:

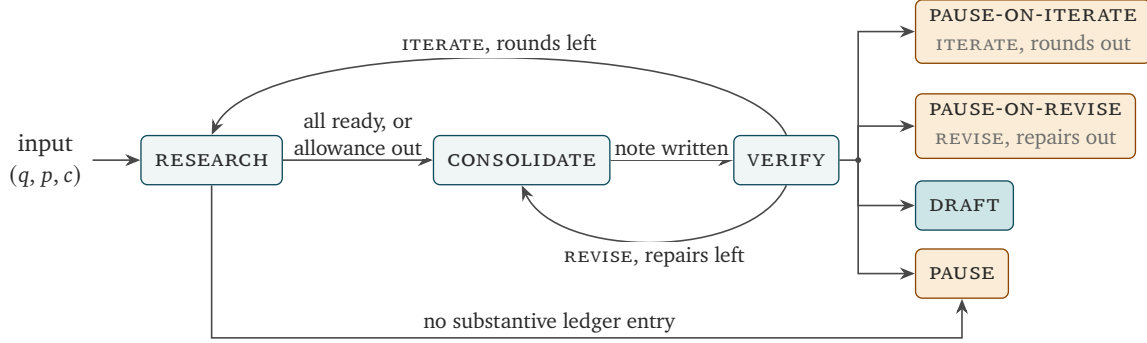
- 1 **DRAFT**: the note carries a supported result that would justify writing a paper.
- 2 **PAUSE**: the investigation does not justify further effort now, or needs inputs the agents cannot supply.
- 3 **PAUSE-ON-ITERATE**: the verifier requests further research after the research-round allowance has been used.
- 4 **PAUSE-ON-REVISE**: the verifier requests another repair after the repair allowance has been used.

An operational interruption is not one of these scientific outcomes.

Denote this terminal set by

$$T_R = \{\text{DRAFT}, \text{PAUSE}, \text{PAUSE-ON-ITERATE}, \text{PAUSE-ON-REVISE}\}.$$

Admission of (q, p, c) initialises M with this input, the role assignments and an empty ledger, sets $\rho = \text{RESEARCH}$, and sets $r = 1, u = w = e = 0$. Candidate selection precedes this research position; it is not another state of the research machine.



Research phase. The input (q, p, c) carries an optional seed connexion c .

Research. Upon receiving a tuple, the runner creates a directory for this investigation, stores the papers and their metadata, and makes the seed available. It launches the researcher agents concurrently, each with a prompt, tools, an allowance and its own working directory. Each sees the papers, seed, whole ledger and its partners' working files. It may write freely in its own directory; shared findings, objections, corrections and intentions go into the ledger. The prompt asks whether there is something publishable to say about the pair, not for a quota of positive results. A continuation also exposes the verifier's feedback.

There is no turn-taking. An agent declares itself ready with the number of the latest ledger entry it has read. Any subsequent substantive entry invalidates that declaration. A researcher whose readiness still holds waits rather than spending another call to repeat itself. The round ends when every researcher has a valid readiness declaration, or when the available calls or time have been used. Readiness need not mean agreement: unresolved disagreements must remain explicit. If no substantive entry has been produced, the thread ends in **PAUSE**; otherwise it proceeds to consolidation.

Writing a ledger entry is a state change even though it does not move the control position. For a permitted entry z by researcher a ,

$$(\sigma, M, b) \xrightarrow{\text{append}(a, z)} (\sigma, M \oplus \text{entry}(a, z), b), \quad \rho = \text{RESEARCH}.$$

The update assigns the next ledger sequence number. A substantive entry invalidates any readiness declaration preceding it; a declaration records the position its author reports having read, not proof of comprehension. Let $\text{Ready}(M)$ mean that every researcher holds a valid declaration, $\text{HasWork}(M)$ that the ledger has a substantive

entry, and $\text{Out}(b)$ that the round's call or time allowance has ended. At a runnable stage boundary,

$$\begin{aligned} \rho &= \text{RESEARCH}, & \text{Ready}(M) \vee \text{Out}(b), \\ (\sigma, M, b) &\xrightarrow{\text{round closed}} (\sigma[\rho \leftarrow \rho'], M \oplus \text{closure}, b), \\ \rho' &= \begin{cases} \text{CONSOLIDATE}, & \text{HasWork}(M), \\ \text{PAUSE}, & \text{otherwise.} \end{cases} \end{aligned}$$

The closure record states whether readiness or exhaustion ended the round. Usage from calls that have returned is already included in b .

Consolidate. The first researcher, an arbitrary but fixed choice, receives a new prompt and writes the research note. It sees the ledger, the researchers' files, and any previous note. Every claim must point to the ledger entries supporting it. The note records what was asked, what holds, the objections and their disposition, and what remains open. If the allowance ended before common readiness, it must not manufacture agreement. On later research rounds it retains the earlier results that still stand and incorporates the new work. On a repair it instead addresses the verifier's corrections without opening a new line of research. Writing the note enables **VERIFY**.

For a returned, structurally usable synthesis n , this is the rule

$$(\sigma, M, b) \xrightarrow{\text{note written}(n)} (\sigma[\rho \leftarrow \text{VERIFY}], M_n, b_n), \quad \rho = \text{CONSOLIDATE}.$$

Here M_n contains the synthesis and its evidence references. Usability means that the required artifact exists and can be supplied to the verifier, not that its scientific claims have already been accepted.

Verify. A verifier that has taken no part in the research is called without tools. The papers, ledger and note are supplied inline; evidence left only in a private working file is not thereby available to this judge. Its reply contains a decision and a reason, with an action when more work is requested:

- **DRAFT:** a supported, non-obvious result warrants writing up.
- **REVISE:** the ledger supports the result but the note misstates it; the reply lists corrections that can be made from the existing record.
- **ITERATE:** a promising direction has a specific gap the researchers can address with their papers, derivations, tools and accessible literature; the reply names the next task.
- **PAUSE:** no supported direction warrants more work now, or progress requires unavailable data, experiments or access; the reply names what is missing.

The verifier must not repeatedly demand an input the record already shows the researchers cannot supply. The runner saves the verdict with a digest of the note judged. A continuing verdict is also added to the ledger as feedback. An unreadable verdict is retained for inspection and leaves the operation **BLOCKED**, not **PAUSE**.

The iterate loop. Research starts at round one. On `ITERATE`, if the research-round cap has not been reached, the runner increments that counter and returns to `RESEARCH`. The researchers keep their ledger and files and address the specified gap. If the cap has been reached, the thread instead ends in `PAUSE-ON-ITERATE`. Thus a request for further research is recorded even when it cannot be executed.

The revise loop. The repair counter starts at zero. On `REVISE`, if a repair remains, the runner increments it and returns to `CONSOLIDATE` with the corrections. The rewritten note is verified again. Otherwise the ending is `PAUSE-ON-REVISE`. Repairs have their own cap because rewriting supported material is different from asking the researchers to investigate again. The reference implementation counts repairs across the investigation; an iterate does not replenish them.

Verifier transition rules. Let ν be a parsed verdict with its reason and, when required, action. Let M_ν save it with the judged note's digest and the feedback required for an executed continuation, and let b_ν include the call's receipt. The complete automatic decision is

$$(\sigma, M, b) \xrightarrow{\nu} (\sigma[\rho \leftarrow \rho_\nu], M_\nu, b'), \quad \rho = \text{VERIFY},$$

with ρ_ν and b' determined by the following table.

Decision in ν	Guard	ρ_ν	b'
DRAFT	true	DRAFT	b_ν
PAUSE	true	PAUSE	b_ν
ITERATE	$r < \bar{r}$	RESEARCH	$b_\nu[r \leftarrow r + 1]$
ITERATE	$r \geq \bar{r}$	PAUSE-ON-ITERATE	b_ν
REVISE	$u < \bar{u}$	CONSOLIDATE	$b_\nu[u \leftarrow u + 1]$
REVISE	$u \geq \bar{u}$	PAUSE-ON-REVISE	b_ν

The guards make explicit that `ITERATE` does not reset u , while `REVISE` does not advance r . No automatic research transition leaves T_R . Output requests can nevertheless act on the other components of σ . An invalid reply does not match this table; the operational failure rule below applies instead.

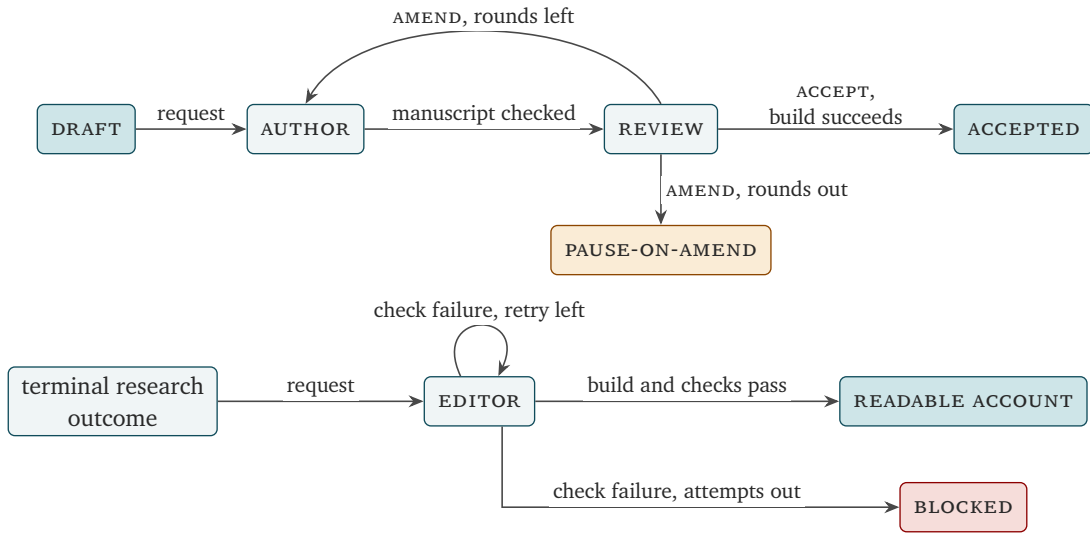
5 The editing phase

Research ends with an assessment of a connexion, not necessarily with a paper ready for a reader. The editing phase has independent output branches. A `DRAFT` enables the *paper* branch. Every terminal research outcome enables the *readable account* branch, whose purpose is to let an expert assess what the research achieved and why it stopped. Neither branch is authorised to change the research record.

Formally, an output request changes only its own component. For a fresh branch, the launch rules are

$$\begin{aligned}
 &\rho = \text{DRAFT}, \quad \pi = \text{IDLE} : \\
 &(\sigma, M, b) \xrightarrow{\text{request paper}} (\sigma[\pi \leftarrow \text{AUTHOR}], M, b[w \leftarrow 1]), \\
 &\rho \in T_R, \quad \eta = \text{IDLE} : \\
 &(\sigma, M, b) \xrightarrow{\text{request account}} (\sigma[\eta \leftarrow \text{EDITOR}], M, b[e \leftarrow 1]).
 \end{aligned}$$

Both require a runnable branch, no stop request, and an available initial attempt. The second rule has no condition on π : an account can precede the paper or be produced when no paper will be requested. Neither request changes ρ .



Editing phase: the paper branch (top) and the readable-account branch (bottom).

Author. The author receives the research note, ledger, verdicts, source papers and researchers' working files. It has tools and web search. Its task is to turn the supported material into a self-contained paper, not to add a theorem that the research record does not establish. It distinguishes the source papers' results, the contribution, interpretation, and limitations. It also searches for prior work on the specific contribution, verifies reference metadata, and records what it searched and found. Discovering that a construction is known changes its attribution and claimed contribution; it is not a reason to conceal the construction or its source.

Check. Once a manuscript and bibliography exist, the runner builds the paper and checks its references. Cited keys must exist; bibliography entries must be cited and carry identifying information. Where an arXiv lookup succeeds, its title is compared with the bibliography. The build result and all findings are passed to the reviewer. A lookup failure is reported as an unavailable check, not as verification of the metadata. A failed build can still be reviewed for correction, but cannot yield an accepted output.

Once a required manuscript exists and its checks have run, the transition is

$$(\sigma, M, b) \xrightarrow{\text{manuscript checked}} (\sigma[\pi \leftarrow \text{REVIEW}], M_c, b_c), \quad \pi = \text{AUTHOR}.$$

The update stores the manuscript, bibliography, search record, checks and author-call receipt. The build need not succeed to enable review: a failed build is evidence the reviewer can use. The check record identifies the version checked so that a later acceptance cannot refer to a different one.

Review. The reviewer is a fresh call without tools. It receives the source papers, research ledger and note, manuscript, bibliography, search record, and check results. It asks whether the paper is fit to send to referees within its stated scope: are its arguments supported and understandable, its attribution fair, and its related-work claims justified by the record? The answer is `ACCEPT` or `AMEND`, with an explanation and identified findings marked major or minor. The runner saves the review with the manuscript's digest.

An `ACCEPT` may include minor corrections to wording or notation that do not change a claim. `AMEND` requests changes to a claim, its scope, attribution or correctness, or another defect that would obstruct assessment. This distinction gives the loop a stopping rule: the target is an assessable manuscript, not an absence of all possible editorial comments. Acceptance with minor findings leaves those findings recorded; it does not imply they have already been applied.

The amend loop. With rounds left, `AMEND` returns the paper to the author with the identified findings. The author addresses each from the existing material and explains what changed. The runner builds and checks the replacement before another review. Each author-reviewer pass counts as a paper round. At the cap, unresolved amendments lead to `PAUSE-ON-AMEND`. If a reviewer accepts a manuscript whose build failed, acceptance is withheld and the remaining correction allowance is used instead. Missing required files or an unreadable review are operational blockages, not scientific rejections.

For a parsed review μ , define $\text{Acceptable}(\mu, M)$ to mean that its decision is `ACCEPT` and the corresponding manuscript has a successful build. Let M_μ save the review, manuscript digest and findings, and b_μ account for the call. Then

$$(\sigma, M, b) \xrightarrow{\mu} (\sigma[\pi \leftarrow \pi_\mu], M_\mu, b'), \quad \pi = \text{REVIEW},$$

where

$$(\pi_\mu, b') = \begin{cases} (\text{ACCEPTED}, b_\mu), & \text{Acceptable}(\mu, M), \\ (\text{AUTHOR}, b_\mu[w \leftarrow w + 1]), & \neg \text{Acceptable}(\mu, M), w < \bar{w}, \\ (\text{PAUSE-ON-AMEND}, b_\mu), & \neg \text{Acceptable}(\mu, M), w \geq \bar{w}. \end{cases}$$

Here the non-acceptable cases are `AMEND` or `ACCEPT` with a failed build, not malformed replies. This rule counts automatic author-reviewer rounds; the owner-directed reviewer-only workflow is described separately.

Readable account. After a terminal research outcome, an editor receives the research record and writes for an expert who has not followed the ledger. It explains the question, approach, supported outcome, unresolved objections, stopping reason and possible next step. Its purpose is assessment of the research, including an unsuccessful attempt, rather than advocacy for publication. It has file tools but no configured web search and is not asked to conduct new research. The runner builds the account and checks its references. A failed build or a blocking reference defect is returned for a bounded correction attempt; failure after the allowance leaves this branch **BLOCKED**. Unavailable external lookups remain disclosed warnings.

The account cannot be produced while the research is still running, **STOPPED** or **BLOCKED**. The research runner normally requests it automatically after a terminal verdict, unless a stop has been requested. The owner can request a deferred account later. It neither waits for paper acceptance nor changes the research verdict, and its own failure does not invalidate the other branch.

Let $\text{AccountOK}(M_c)$ mean a successful build with no blocking reference defect; an unavailable external lookup remains a warning. After an editor attempt and its checks,

$$(\sigma, M, b) \xrightarrow{\text{account checked}} (\sigma', M_c, b'), \quad \eta = \text{EDITOR}, \quad \rho \in T_R,$$

where $\text{AccountOK}(M_c)$ sets $\eta' = \text{READABLE ACCOUNT}$. Otherwise, $e < \bar{e}$ keeps $\eta' = \text{EDITOR}$ and sets $e' = e + 1$; at the cap it keeps the saved editor position and sets $\omega'_A = \text{BLOCKED}$. All other control components are unchanged, and b' includes the attempt's usage. For each editing transition, the research projection of the record obeys

$$M' \upharpoonright_R = M \upharpoonright_R .$$

This projection contains the input, research ledger, synthesis and scientific verdicts, not the output branch's own files or receipts. It states precisely what it means for editing to leave the research outcome alone.

6 The campaign and its operational control

A *thread* is an instance of the machine for an input tuple. A *campaign* supplies the corpora, candidate selection, resource settings and a scheduler for several threads. An *owner*, human or delegated, chooses these settings and decides what to do with outputs and pauses. The scientific transition rules are the same whether a thread is run alone or occupies a seat in a campaign.

Admission. Fetching and scanning precede the machine described here. The scanner assigns a score and possibly a seed connexion to candidate pairs; selection produces a shortlist. A free seat and a passing budget guard permit the runner to admit a shortlisted investigation. Its stored position determines whether to start research or resume an unfinished stage. Terminal threads are not automatically researched again. The corpora and shortlist can grow without restarting threads already admitted. In

the reference layout a campaign and pair identify a thread; another independent attempt needs a separate campaign or record.

Shared memory. The ledger is an append-only sequence of attributed, numbered entries. Appends are serialised, although the researchers work concurrently. Substantive entries carry ideas, findings, objections, corrections or intentions; readiness and review entries have the control roles described above. The record also contains the current notes and manuscripts, judgement histories, saved positions and resource receipts. Saving a judgement with an artifact digest identifies what was judged. A digest does not recover an overwritten version: full replay would additionally require retaining that version and its invocation context.

The contracts say which records each role can use and what it may change. Researchers do not edit their partners' files; the editing branches leave the research record alone; judges receive their evidence rather than acquiring unrecorded evidence with tools. These are role contracts, not a claim that every file permission enforces them. Model and prompt choices remain part of the run's provenance.

Context and ledger access. Ledger freshness is part of the researchers' contract. Each researcher tracks the sequence number k of the latest entry it has read and refreshes its view before a major change of direction and before declaring readiness. A readiness declaration names that observed position; it cannot certify unseen entries. Any substantive entry after k , including one appended between the refresh and the declaration, makes that declaration stale. The runner determines common readiness from the ledger, not from an agent's unqualified statement that it has finished.

Incremental reading is a recommended implementation refinement. A ledger helper may return only entries after k , in sequence order, together with the position reached; the researcher advances its cursor after reading them. This avoids repeatedly supplying material already available in the current invocation. A cursor alone is not a substitute for that material: a fresh invocation must receive the existing ledger before consuming subsequent updates. After context compaction, the agent must retain or reload the earlier evidence needed for its next claim; the cursor does not establish that this evidence remains available. A full refresh is the safe fallback. These rules do not require incremental retrieval in the reference helper.

Prefix sharing is a separate execution optimisation. Within the shared task context, the current implementation places the source papers first, followed by the growing ledger, before caller-specific instructions and changing material. Judges also receive the note being assessed. What one call can reuse from an earlier one is therefore the leading part only, the papers and the ledger entries already present; the new entries, the note and the instruction after them change. An implementation should preserve the common material and its ordering between calls, placing unnecessary per-call variation outside the reusable prefix. The backend must configure any required cache boundaries and report actual cache usage rather than infer a cache hit from shared text. Provider-specific cache settings do not belong to the transition system. Every call must still supply its required context, whether directly or through the backend's

supported context mechanism; correctness and ledger freshness must never depend on a cache hit.

Allowances. The resource settings attach to different loops and should not be conflated.

Allowance	Scope	Exhaustion effect
Researcher calls and time	A research round	Consolidate the existing ledger.
Research rounds	The iterate loop	PAUSE-ON-ITERATE.
Note repairs	The revise loop	PAUSE-ON-REVISE.
Paper rounds	The amend loop	PAUSE-ON-AMEND.
Account attempts	Build and reference repair	The account is BLOCKED .
Campaign expenditure	Admission of threads	Stop new admission.

Each model call also has a deadline. The reference defaults allow four research rounds, one note repair, three paper rounds and two account attempts; these numbers are settings, not properties of the scientific task. Within a research round the call cap applies per researcher and elapsed call time is accumulated across researchers.

Every attempted call appends a receipt, specified below. The admission guard compares recorded expenditure, a per-call estimate for every call that established a session and whose cost is unknown, and a reservation for work in flight against the campaign budget. This is a scheduling guard, not a strict upper bound on the eventual bill: admitted threads can need further calls, and concurrent researchers can overshoot a shared time allowance before their usage is collected. Token usage is recorded, but the present runner does not enforce a separate hard token quota. Unpriced subscription usage must not be interpreted as free work. Strict reservation before every call would require a stronger accounting mechanism.

Call receipts. The receipts are the campaign's only record of what it consumed, and every statement about cost or time rests on them. They are therefore part of the specification. The runner appends one receipt for every call it attempts, including an agent that could not be launched, a call that never established a session, and a call ended at its deadline. This covers the attempts the runner handles; it is not a promise of durable accounting across an abrupt crash of the runner itself.

A receipt records the time, the thread, the stage and the role, the backend and the model requested, the elapsed time, and an outcome: completed, error, timeout, no session, or launch failed. It records the provider's usage counters as the provider reported them, with the backend named, because providers count differently: one includes cached tokens in its input total where another reports them apart, and reasoning may or may not be included in output. A breakdown into uncached input, cache reads, cache writes and billed output is derived later from the raw counters and their documented meaning; it is not substituted for them.

What was not reported stays unknown. A missing counter is recorded as absent, never as zero, while a reported zero remains zero. Counters that did arrive are kept

on a failed or interrupted call as well, where the outcome tells the reader that they may be partial: such a call can consume resources without saying how much. A cost, in US dollars, carries its basis. It is either the amount the provider reported, or an amount priced from reported counters at rates the receipt retains, which is an approximation and not a reconstructed provider charge. When neither is possible the cost is unknown. An amount computed for a subscription is an equivalent, not an additional bill. No receipt holds an estimate: the guard's caution about calls of unknown cost belongs to the guard, which does not charge a call that never reached a session, and a display of expenditure states the known cost and, separately, how many calls have unknown cost. Repricing under another tariff is an analysis over receipts and adds nothing to them.

The guarantee is this: enough is recorded to reconstruct what a campaign consumed and, where the provider reports enough, to reprice it, and what cannot be known is visibly unknown. The reference receipts do not identify retries; that identity must not be inferred from adjacent receipts, since an ordinary later call can carry the same thread, stage and role.

Stop. A stop request prevents new admission and is tested at call or stage boundaries. Calls already in flight are allowed to return, and their work is recorded. A thread that has not reached a terminal verdict keeps its current stage and becomes STOPPED. Clearing the request permits continuation from that stage. It does not require finishing the whole investigation before the runner stops, and it does not turn an interruption into PAUSE.

The operational mode makes this distinction explicit. For a branch j that still has work, at a safe boundary,

$$(\sigma, M, b) \xrightarrow{\text{stop boundary}_j} (\sigma[\omega_j \leftarrow \text{STOPPED}], M, b).$$

The stage positions ρ, π, η are unchanged. No new call is launched while its branch is not runnable or a stop request is pending. An in-flight call may still return and record its result and usage; the stop guard is applied before launching further work. An account request is therefore deferred, not enabled merely because research was interrupted.

Failure and recovery. Failure to establish a model session is a transport failure: the scientific decision has not been made. The first operational failure observed by the research scheduler stops campaign admission. Active work drains before a nonzero exit; there is no automatic health-probe loop. Editor failures propagate as operational failures, and terminal research with an unfinished account remains pending for an explicit restart. A call that starts but times out may already have written useful files. Recovery therefore considers both the saved position and the required artifacts, rather than treating every timeout as a fresh investigation. Bounded call retries are distinct from the verifier's requests for new research.

A blocked operation has no usable contracted output, for example an unreadable verdict, and waits for inspection. Reconciliation chooses the next operational step: resume research, consolidate, or verify an existing note. It does not invent a scientific

verdict. The implementation saves files and positions separately, so a crash between them can require repeating a call. Researcher time and call counters within a round are not durable across every restart. The guarantee is preservation of recorded work, not exactly-once execution or an unchanged remaining allowance after arbitrary failures.

An external supervising agent can audit the runner every five minutes through `pathfinder health -json`. The read-only snapshot separates scheduler heartbeat from completed-investigation progress and lists active call identities, owner and child process identifiers, nominal deadlines, and recent failures. Overdue work or a stale heartbeat warrants investigation; neither proves a deadlock. Existing transport startup and cleanup can exceed the nominal call allowance. An exclusive operating-system lock prevents concurrent research runners, and restart refuses recorded calls whose owner or child process remains alive. Process identity and surviving descendants still require inspection. The supervising agent owns scheduling, diagnosis, and an incident ledger of evidence, action, and subsequent progress. The package does not itself install that recurring audit or guarantee delivery to an agent.

For example, a returned but unparsable judge reply z has the rule

$$(\sigma, M, b) \xrightarrow{\text{bad reply}_j(z)} (\sigma[\omega_j \leftarrow \text{BLOCKED}], M_z, b_z).$$

It records the raw reply and usage but supplies no `DRAFT`, `PAUSE`, `ACCEPT`, or `AMEND` label. A transport failure instead marks the unfinished branch `STOPPED` and records the failure. A resume or reconciliation event can restore `RUNNABLE` only after the stop or failure condition has been addressed and the saved artifacts identify a usable continuation. These operational events do not manufacture a scientific verdict or silently start a new independent investigation.

Owner intervention. A terminal research outcome ends that automatic run. Raising an allowance alone does not reopen it in the reference implementation. The owner can supply missing evidence and start a separately recorded investigation, or arrange an explicit continuation while retaining the old outcome. For a paper paused on amendments, increasing the paper-round cap permits further author–reviewer passes. Alternatively, the owner can edit it and request a reviewer-only pass; this counts as another review without an author call and can exceed the automatic round cap. These manual calls still consume resources, even though the admission guard does not bound that workflow.

The runner uses a per-thread ownership marker to avoid ordinary duplicate admission. Independent paper or editor commands must also be serialised by the owner; the reference runner’s marker is not a general transactional lock covering every command. A stronger implementation can enforce these ownership rules without changing the scientific states.

7 What the process establishes

The machine separates proposing a connexion, supporting it, and presenting it for assessment. **DRAFT** means that the verifier found enough supported material to justify writing. **ACCEPTED** means that the internal paper reviewer found the manuscript assessable within its stated scope. Neither label certifies scientific truth, exhaustive novelty, or eventual acceptance by a journal. Conversely, a pause can be informative: its account should identify a failed construction, a missing input, or a precise limit on what the current materials support.

Independent calls provide a separation of roles, not independence of every model error. A shared misconception can pass through researchers and judges. Correctness, novelty and significance remain questions for the expert reader, who can inspect the sources, arguments and limitations. This is why the readable account is an output in its own right, rather than merely a by-product of successful paper writing.

An implementation of this description should preserve the guarded transitions, the evidence available to each role, the distinction between scientific and operational endings, and the records needed to explain each decision. It need not reproduce the same model responses. Pathfinder organises attempts at scientific discovery; it does not make their outcome deterministic.

The notation also exposes useful invariants of the automatic machine:

$$\begin{aligned} \eta \in \{\text{EDITOR, READABLE ACCOUNT}\}; &\implies; \rho \in T_R, \\ \pi \neq \text{IDLE}; &\implies; \rho = \text{DRAFT}, \\ \pi = \text{ACCEPTED}; &\implies; \text{an accepting review and successful build} \\ &\text{identify the same manuscript version.} \end{aligned}$$

They follow from the launch and completion guards, preservation of the research record by editing, and the absence of an automatic transition out of a terminal research position. Stops and blockages change operational modes, not these scientific prerequisites. Explicit owner continuations must preserve their prior outcomes and provenance rather than be confused with an automatic edge in the original run.

A Agent resources against human resources, per pair

The economics has several distinct measures: tokens consumed, model charges, elapsed time, and human attention. The root campaign used 330 model calls with 186.58 USD of recorded charges; its outcomes and mixed-model setup are described in Section 2. Later reruns and repairs are excluded. The recorded activity spans 11–13 September, including interruptions and owner-directed continuations, rather than an uninterrupted benchmark run.

Agent resources. Table 4 carries over the approximate per-stage figures from the original campaign summary. That summary reports typical costs and times as medians

over threads, with the qualifications shown below. These figures have not been recomputed here from individual call intervals and should not be added to obtain a measured end-to-end median. Dollar amounts use the historical accounting rates, not a common GPT-6 price basis.

Stage, per pair	Calls	USD	Elapsed time
Scan	1	0.03	11 s
Research, consolidation and verification	12	6.8	8 min
Manuscript preparation and review	4–11	3.5	5 min per round
Readable account	1	1.0	2 min

Table 4 Approximate figures retained from the original campaign summary. Research was reported to average 9.8 USD and 51 minutes, with a worst thread around 30 USD and six hours. The longest manuscript sequence used seven review rounds, with a reported cost of 14.7 USD. These are descriptive figures from a development campaign, not controlled performance estimates.

Tokens and a common price basis. The original summary does not give token totals, and dollar amounts cannot be converted back into token counts without their breakdown. Repricing a campaign requires separate counts for uncached input, cache reads, cache writes, and billed output including reasoning. GPT-6 Astra’s standard API rates on 16 September 2026 are respectively 10, 1, 12.50, and 50 USD per million tokens for requests of at most 272,000 input tokens; above that threshold the corresponding rates are 20, 2, 25, and 75 USD for the full request.⁴ A common-price comparison would apply those rates to the recorded token categories, retaining any separately billed tools. It would be a *GPT-6-equivalent token cost*, not an observed GPT-6 run. Choosing medium reasoning effort does not determine how many tokens a rerun would consume. Subscription charges and quota usage are separate measures, not zero-cost API usage. No normalised campaign total is asserted here.

An illustrative human comparison. Table 5 retains the original note’s estimates for a postdoctoral-level researcher, valued at an assumed fully loaded rate of 60 USD per hour. These are workload assumptions, not observations from a human control group or claims about prevailing salaries. They describe similar stages of work, not demonstrated equality of scientific quality.

Under the scanning assumption alone, examining 100 pairs takes 25 human hours, compared with 3.34 USD of recorded scan charges. This illustrates why broad searches become affordable; it does not establish a quality-matched speedup. A human researcher would normally select promising papers rather than inspect every pair, and that selection is itself part of the expertise being compared.

Human effort does not disappear from an agent campaign. The original account estimated 10–20 minutes of owner attention per pair, plus about an hour for a

⁴ OpenAI’s published pricing and context threshold: <https://developers.openai.com/api/docs/models/gpt-6-astra> and <https://developers.openai.com/api/docs/pricing>.

Human activity, per pair	Hours	USD at 60/h
Read two abstracts and decide	0.25	15
Read both papers, investigate and write a research note	25–40	1,500–2,400
Colleague checks the research note	2–4	120–240
Prepare a paper with related-work search	60–80	3,600–4,800
Review the paper	4–8	240–480
Write a readable account	4	240

Table 5 Illustrative human effort from the original note. A total would require deciding how much research, literature search and writing overlap between stages. Calendar time also depends on availability and review delays.

manuscript requiring manual intervention; this was not a time-tracked measurement. Reading and checking the scientific outputs remain necessary. The potential saving is in moving human effort from producing every candidate to selecting, assessing and developing promising ones, not in replacing scientific assessment with internal acceptance.

Finally, elapsed time is not the sum of agent call durations. Researchers overlap within a round, stages follow one another, and several threads may run concurrently. Timeouts, queueing and pauses can dominate the calendar. Cost, active computation time, campaign elapsed time and human attention should therefore be reported separately. Neither the historical timing figures nor the human estimates justify a general multiplier for speed or cost advantage at equal scientific quality.

B Conformance tests for a reimplementation

Scientific outputs cannot check a reimplementation, because the model responses are not deterministic and need not agree between runs. What is checkable is the transition system: the guards, the counters and their caps, the terminal labels, and the separation of operational from scientific endings. These can be tested without spending model calls.

The reference implementation does so with a substitute agent command driven by environment variables: what to reply, how long to wait before announcing a session, whether to hang instead of replying, and a shell command to run in the working directory before replying. The substitute must write every file a real call writes. Otherwise a branch that real calls never reach will pass.

The following paths exercise the specification above, and a reimplementation that passes them may be considered faithful to it.

- A scan interrupted and resumed, and an unparsable scored row retrieved.
- Selection by percentage cut, and by threshold with already started investigations retained.

- A thread reaching each of DRAFT, PAUSE, PAUSE-ON-ITERATE and PAUSE-ON-REVISE.
- The iterate cap, reached with the verifier still requesting research.
- A repair applied once and the repair cap reached thereafter.
- An empty ledger ending the thread in PAUSE.
- A stop request draining a call in flight, and a resume continuing from the recorded stage.
- The admission guard writing the stop marker when the budget is reached.
- An operational failure setting the health marker and stopping admission without probing, even when another investigation succeeds.
- Terminal research resuming unfinished editing without repeating research.
- Read-only health snapshots distinguishing stale heartbeats, missing runner processes, and overdue calls, with exclusive ownership released on process death.
- An agent that cannot be launched, and a call that never establishes a session, each leaving a receipt with no usage and no cost.
- A call ended at its deadline leaving a receipt that keeps whatever usage was reported, and none when none was, with the guard still counting it; and a missing counter recorded as unknown while a reported zero stays zero.
- Both blocked paths, an unreadable verdict and an unreadable review, and reconciliation naming and applying the one safe action.
- A killed runner leaving an ownership marker whose process is gone.
- Append-only paging in open-ended mode, with existing pair identifiers unchanged.
- The paper and readable-account round trips against a real LaTeX build.

The invariants stated at the end of Section 7 can be asserted directly over the saved records while these tests run.

After they pass, a pilot on two corpora of about five papers a side, with stop, resume and reconciliation exercised by hand, establishes what a campaign costs and how long it takes before a larger one is attempted.

C Correspondence with the reference implementation

The following names connect the description to the repository without making the directory layout part of the mathematical state space. The pair identifier is local to a campaign. Each role's prompt supplies the scientific criteria described in the corresponding transition paragraph.

Record or operation	Reference representation
Input papers and metadata	threads/<pair>/inputs/; the optional seed is taken from the campaign scan.
Research record	ledger.jsonl, researcher directories and <pair>.tex.
Research position and judgements	status.json and <pair>.verdict.json.
Paper branch	paper/, including the manuscript, bibliography, search record, checks and reviews.
Readable account	edited/, with its source, bibliography, PDF and attempt status.
Campaign control	Shortlist, configuration, receipts, stop marker and health marker.
Role instructions	peer.md, consolidate.md, verify.md, author.md, review.md, editor.md.

The verifier’s structured reply has fields decision, reason and action; the last carries corrections or the next research task, and is otherwise empty. The paper reviewer returns decision, summary and findings; each finding has an identifier, severity, location, issue and proposed fix. The runner, rather than the judge, adds counters, timestamps and artifact digests. State labels are rendered uniformly in this note; their capitalisation and file-field names in the repository are serialization details.

D Variability, evaluation and discovery comparison

Repeating an investigation can produce a different connexion or a different verdict. Moreover, judging the resulting science is itself uncertain. These are separate questions: how variable is the research, and how reliably can we assess it? The following related work motivates a proposed experiment, not a further result of the pilot.

D.1 What existing evaluations measure

Tasks with assessable answers. The starting questions in the studies below come from benchmark instances. There is no research-seed generation stage to compare with Pathfinder’s scan. Their agents generate candidate answers or action plans, not new research questions. *More Agents Is All You Need* studies sampling and voting on benchmark tasks; its sampling-parameter robustness experiments average over twenty runs [5]. *Debate or Vote* compares independent voting with agent discussion against benchmark answers. Its main table reports five-run variation for the single-agent baseline, but not equivalent error bars for the multi-agent configurations [6]. The broader comparison by Kim et al. uses task completion and correctness across agent architectures, with confidence intervals and robustness analyses [7]. Wunderlich et al. compare accuracy at comparable computational cost, reporting sampling uncertainty over benchmark questions [8]; Tran and Kiela likewise show why controlling the reasoning budget matters [9]. In contrast, τ -bench explicitly repeats

tasks and distinguishes finding a successful attempt from succeeding consistently across attempts. Its tasks are constructed from human-designed domains, policies and annotated user requests, with model assistance for data generation. Evaluation relies on an annotated target database state [10], an advantage unavailable for an undiscovered result.

Classifying scientific-research systems. The closer comparisons concern systems that propose research rather than solve supplied benchmark tasks. We distinguish how the starting idea is generated, the protocol that develops it, the output produced, and how that output is evaluated. In describing the starting idea, supplying a broad topic is different from supplying the hypothesis to investigate. Likewise, review inside the generation protocol is distinct from evaluation of the system by its authors or independent readers.

VirSci [11]. *Seed generation:* agents represent scientists with publication-based profiles. A sampled leader recruits collaborators, and the team chooses a topic through discussion, then retrieves relevant papers and proposes ideas. *Generation protocol:* agents take turns proposing or revising ideas and experimental plans, retain highly self-rated candidates, and vote on novelty against retrieved literature. They iteratively refine the selected idea’s abstract. *Output:* a research abstract describing proposed work and expected results, not an executed study. *Evaluation:* literature-similarity metrics, agent reviews and human reviews. Scenarios are repeated twenty times; a 200-abstract validation uses three human assessments and three agent assessments per abstract.

Si et al. [12]. *Seed generation:* a human-supplied topic guides literature retrieval. The agent generates 4,000 candidate ideas per topic, using example proposals and, for some generations, retrieved titles and abstracts. *Generation protocol:* generate many candidates, remove near-duplicates, expand ideas into structured project proposals, and rank them through pairwise agent comparisons. A separate condition uses human reranking. *Output:* research proposals, not completed investigations. *Evaluation:* blind expert comparison with human-written proposals, using novelty, feasibility, expected effectiveness and related criteria; the analysis accounts for topic, idea and reviewer variation and measures reviewer disagreement.

The AI Scientist-v2 [13]. *Seed generation:* humans supply broad topical prompts, chosen for the workshop’s scope in the reported experiment. The system formulates hypotheses and experimental plans while consulting literature through Semantic Scholar. The topic is human-provided; the specific research idea is agent-generated. *Generation protocol:* an experiment manager coordinates code generation, execution and refinement through tree search, progressing from a working prototype through tuning, the research agenda and ablations. Selected experiments are replicated; the system generates figures and a manuscript, with model feedback and revision. No supplied experimental code template is required. *Output:* full experimental papers. *Evaluation:* internal model feedback, author inspection and external workshop

peer review. The authors submitted three manuscripts and report that one met the acceptance threshold.

Pathfinder. *Seed generation:* humans select corpora Q and P . The scanner examines pairs in $Q \times P$, proposes connexions from titles and abstracts, and scores whether they warrant investigation. The specific question emerges from this encounter rather than from a supplied topic to elaborate. *Generation protocol:* researchers investigate with a shared ledger; consolidation and verification lead to bounded cycles of further research or repair, or a terminal outcome. Researchers may abandon or reformulate the seed. Editing follows research, producing a reviewed manuscript when warranted and a readable account of terminal research outcomes. *Output:* a research record and, where justified, a pre-publication paper; paused investigations also yield an account of what was learned. *Evaluation:* internal agent verification and manuscript review, plus build checks. The pilot inventories resulting claims but does not establish an independently validated discovery rate. The experiments proposed below would compare scientific contributions across protocols.

Generating ideas and producing papers both have precedents. Pathfinder’s distinguishing design choice in this comparison is its systematic use of cross-corpus pairings to generate and select questions, making the candidate space and its coverage explicit. Other systems also use prior literature; the distinction is not simply “ideas from papers”. This identifies a design difference, not a claim of priority over all literature-based discovery methods. In every case, assessment of scientific claims remains necessary beyond the output format or an acceptance decision.

D.2 Independent research runs, compared afterwards

Rather than ask reviewers to give every output an overall quality score, we could compare the specific contributions found by different protocols. The labels would be constructed afterwards from all protocols’ findings, not fixed in advance or supplied by the current protocol alone. This experiment stops after the research phase, including consolidation and verification: neither the manuscript nor the readable-account editing branch is needed.

- 1 **Run independently.** Repeat each protocol on fixed example pairs, initially Q3P3 and Q4P6. Compare the current protocol with alternatives such as a single researcher, independent researchers followed by consolidation, and interacting researchers. Fix the papers, optional seed, model, tool access, resource allowances and number of runs before examining the outcomes. Retain the same final verification procedure and include coordination and verification in the budget. Keep every research record, including pauses and operational failures. No run receives another run’s findings; access to published pilot outputs must also be controlled and documented.
- 2 **Construct the labels.** Pool the completed research records and have assessors, blind to the producing protocol, identify and check their contributions. Each label specifies a claim, its assumptions, its supporting argument or experiment,

and the criterion for an equivalent finding. Merge duplicate formulations without merging substantively different assumptions or guarantees. Fix the labelling rules in advance, although the labels themselves emerge afterwards. Double-assess a subset to measure disagreement.

- 3 Compare the findings.** Map each run to its checked labels, then report shared and protocol-specific contributions, their frequency across repeated runs, and the number of distinct contributions found at comparable total expenditure. Retain the evidence for each attribution: mentioning a conclusion is not equivalent to establishing it.

For a pair x , let $L_{\pi,j}(x)$ be the checked labels found by run j of protocol π , and define the pooled inventory and each run's coverage by

$$L_x = \bigcup_{\pi,j} L_{\pi,j}(x), \quad \text{coverage}_{\pi,j}(x) = 100 \frac{|L_{\pi,j}(x)|}{|L_x|} \quad \text{when } |L_x| > 0.$$

If no checked contribution is found for a pair, report that outcome rather than an undefined percentage. Give per-pair coverage distributions and per-label frequencies, not just the best run. A launch without a research output contributes no labels and is also recorded as an operational failure. Report actual tokens, cost and elapsed time alongside the scientific outcomes.

The examples in Section 2 and the talk's inventory illustrate possible labels: the conditional judge-free ranking construction for Q4P6, or the identification and design of a missing matched-budget control for Q3P3. The latter is an experimental-design contribution, not evidence that the critic is ineffective. Keep proposed tests distinct from proved or experimentally established results. Supported obstructions may count even when a run pauses; internal DRAFT status alone establishes neither validity nor novelty.

What this comparison would establish. Coverage measures the share of the pooled findings, not the share of all possible discoveries. The inventory depends on the protocols and runs sampled; the comparison is therefore exploratory, not a held-out test against an independent answer key. It allows a new contribution from any protocol to enter on equal terms, while distinguishing repeated access to familiar findings from complementary discoveries. A protocol-specific label is not automatically more valuable, and counts depend on the agreed granularity. Record unsupported claims separately so that high coverage cannot conceal unreliable research. The inventory could subsequently be frozen for a fresh recovery experiment. Neither comparison establishes generalisation beyond the tested pairs, nor does evaluation before editing measure manuscript quality. It makes scientific assessment more specific, not unnecessary.

D.3 Virtual conferences and successive generations

A more ambitious experiment would turn each campaign into a virtual conference. Starting from the same source corpora, each protocol produces a batch of research

papers, submits them to a common review procedure, and returns admitted papers to the pool of candidates for subsequent scans. Further generations can then investigate connexions involving earlier generated work. Fix an admission and pairing rule in advance: which papers enter Q , P , or both, and whether generated papers may be paired with each other. Preserve source versions and ancestry throughout. Admission is permission to investigate a paper further, not certification that its claims are true.

This asks whether a protocol can sustain cumulative research, rather than recover a fixed inventory. For example, a later paper might relax the assumptions of the Q4P6 ranking construction, establish its finite-sample requirements, or find a counterexample. These are possible directions, not results already obtained. A contribution must add something beyond both the initial literature and its generated predecessors; a new pairing, title or restatement is not enough. Corrections and supported negative results also count, but an unsupported premise inherited from an earlier paper must not silently become an established fact.

Compare separate protocol lineages from the same initial corpora, with the same review procedure and resource allowance per generation, and repeat each lineage independently. Count distinct, independently checked contributions and their evidential types, alongside invalidated claims, costs and human assessment effort. Plot cumulative contributions against total expenditure, not merely against generation number or paper count. Keep the assessors separate from the producing agents; agent review may screen candidates, but human audits and claim-specific checks remain necessary. If only a sample is audited, report that sampling and its uncertainty rather than describing all admitted papers as validated.

The evolving corpora are themselves an outcome: a successful discovery may make later progress easier. Such a comparison measures a whole research lineage, including selection and accumulated opportunities. To distinguish this effect from a better research procedure on the same inputs, periodically test the protocols on a common frozen pool. Keep this longitudinal evaluation separate from the single-generation comparison: its aim is to measure additional supported ideas, not agreement with an ever-expanding answer key.

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